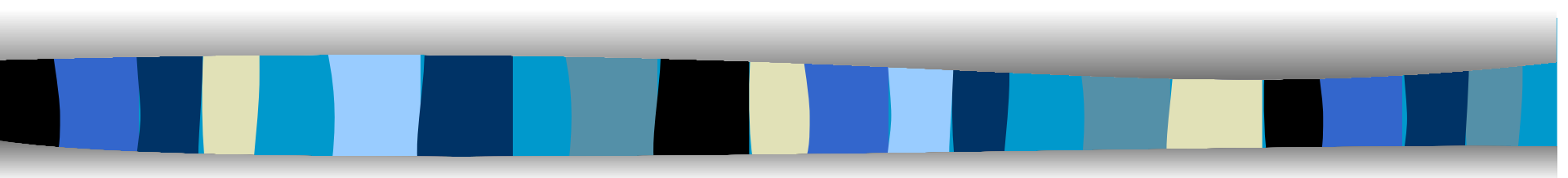


Asymmetric Cryptography

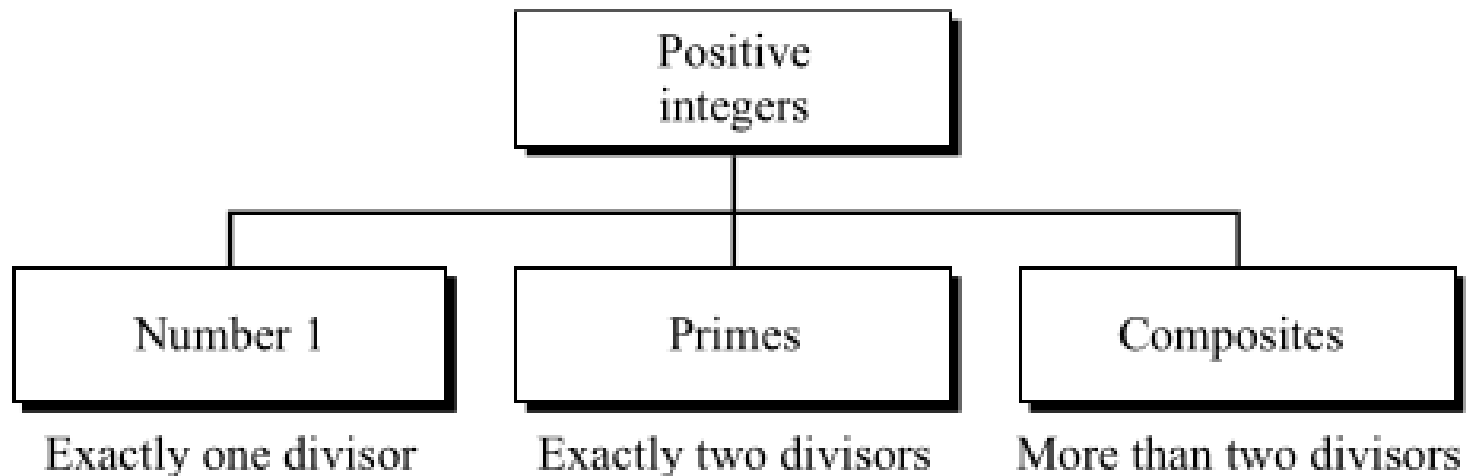


Number Theory

Primes

- Asymmetric-key cryptography uses primes extensively.
- A prime number is divisible by itself or by 1.

Three groups of positive integers



Co-primes

- Two positive integers, a and b , are relatively prime, or coprime, if $\gcd(a, b) = 1$.
- Note that the number 1 is relatively prime with any integer.
- If p is a prime, then all integers 1 to $p - 1$ are relatively prime to p .
- Z_{n^*} is the set of integers less than n which are all relatively prime to n .
- Set Z_{p^*} is the same except that modulus (p) is a prime.

Cardinality of prime

- There are infinite number of primes.
- So, cardinality of primes is infinite.

Checking primeness

- Check if n is prime
- $q = \text{floor}(\text{sqrt}(n))$
- Check if any primes less than q divides n
- If so, then n is not a prime
- Else, n is prime.
- Check if 97 is prime
- $q = \text{floor}(\text{sqrt}(97)) = 9$
- The primes less than 9 are 2, 3, 5, 7
- None of these divides 97. So, 97 is prime.

Euler's phi function

- Euler's phi-function, $\varphi(n)$, also known as the Euler's totient function.
- It has a very important role in cryptography.
- The function finds the number of integers that are both smaller than n and relatively prime to n .
- So, it helps to find cardinality of set Z_{n^*}
- The function $\varphi(n)$ is defined as

Euler's phi function

- $\varphi(1) = 0$.
- $\varphi(p) = p - 1$ if p is a prime.
- $\varphi(m \times n) = \varphi(m) \times \varphi(n)$ if m and n are relatively prime.
- $\varphi(p^a) = p^a - p^{a-1}$ if p is a prime and a is some integer
- $\varphi(p_1^a * p_2^b) = (p_1^a - p_1^{a-1}) * (p_2^b - p_2^{b-1})$

Euler's phi function

- $\varphi(13)=13-1=12$ $Z_{13^*}=\{1,2,3,4,5,6,7,8,9,10,11,12\}$
- $\varphi(10)=\varphi(2*5)=\varphi(2)*\varphi(5)=(2-1)*(5-1)=4$
- $Z_{10^*}=\{1,3,7,9\}$
- $\varphi(240)=\varphi(24*10)=\varphi(2^3*3*2*5)=\varphi(2^4*3*5)$
- $= (2^4-2^{4-1})*(3^1-3^{1-1})*(5^1-5^{1-1})=(16-8)*2*4=64$

Format's little theorem

- There are two versions of Format's little theorem.
- The first version says that if p is a prime and a is a positive integer such that p does not divide a , then
- $a^{p-1} \equiv 1 \pmod{p}$.
- This is true for all positive integers $a \neq p$

Format's little theorem

- Let $p=5$
- $1^4 \bmod 5=1$
- $2^4 \bmod 5=1$
- $3^4 \bmod 5=1$
- $4^4 \bmod 5=1$
- $6^4 \bmod 5=1$
- $7^4 \bmod 5=1$
- $8^4 \bmod 5=1$

Format's little theorem

- Let $p=7$
- $1^6 \bmod 7=1$
- $2^6 \bmod 7=1$
- $3^6 \bmod 7=1$
- $4^6 \bmod 7=1$
- $6^6 \bmod 7=1$
- $8^6 \bmod 7=1$
- $9^6 \bmod 7=1$

Format's little theorem

- Another version is
- $a^p \equiv a \pmod{p}$.
- Where p is prime, a is positive integer $a < p$

Format's little theorem

- Let $p=5$
- $1^5 \bmod 5=1$
- $2^5 \bmod 5=2$
- $3^5 \bmod 5=3$
- $4^5 \bmod 5=4$

Format's little theorem

- Let $p=7$
- $1^7 \bmod 7=1$
- $2^7 \bmod 7=2$
- $3^7 \bmod 7=3$
- $4^7 \bmod 7=4$
- $5^7 \bmod 7=5$
- $6^7 \bmod 7=6$

Format's little theorem

- $6^{10} \bmod 11 = 1$
- Find $3^{12} \bmod 11$
- $= 3^{11} * 3 \bmod 11$
- $= [(3^{11} \bmod 11) * (3 \bmod 11)] \bmod 11$
- $= [3 * 3] \bmod 11$
- $= 9$

Inverse using Fermat's little theorem

- We can find multiplicative inverse using Fermat's little theorem.
- $a \times a^{-1} \bmod p$
- $= a \times a^{-1} \times 1 \bmod p$
- $= a \times a^{-1} \times a^{p-1} \bmod p$
- $= a \times a^{p-2} \bmod p$
- $= a^{p-1} \bmod p$
- $= 1$

Inverse using Format's little theorem

- Find the 8^{-1} modulo 17.
- $8^{-1} \bmod 17$
- $=8^{17-2} \bmod 17$
- $=8^{15} \bmod 17$, $a=8$, $m=15$, $n=17$, $z=1$
- Here, 15 in binary is 1111

i	B_i of m	$z=z^2 \bmod n$	$z=z*a \bmod n$
3	1	1	$=1*8 \bmod 17=8$
2	1	$8^2 \bmod 17=13$	$=13*8 \bmod 17=2$
1	1	$2^2 \bmod 17 = 4$	$=4 *8 \bmod 17 =15$
0	1	$15^2 \bmod 17 =4$	$=4 *8 \bmod 17 =15$

Inverse using Fermat's little theorem

- Find the 5^{-1} modulo 23.
- 5^{-1} modulo 23.
- $=5^{23-2} \bmod 23$
- $=5^{21} \bmod 23$
- 21 in binary is 10101 which is 16, 4, 1
- $5^{21} \bmod 23 = 5^{16} \times 5^4 \times 5 \bmod 23$
- $5^2 \bmod 23 = 2$
- $5^4 \bmod 23 = (5^2)^2 \bmod 23 = (5^2 \bmod 23)^2 \bmod 23$
- $= 2^2 \bmod 23 = 4$

Inverse using Fermat's little theorem

- Find the 5^{-1} modulo 23.
- $5^8 \bmod 23 = (5^4)^2 \bmod 23 = (5^4 \bmod 23)^2 \bmod 23$
- $= 4^2 \bmod 23 = 16$
- $5^{16} \bmod 23 = (5^8)^2 \bmod 23 = (5^8 \bmod 23)^2 \bmod 23 = 16^2 \bmod 23 = 3$
- Now, $5^{21} \bmod 23 = 5^{16} \times 5^4 \times 5 \bmod 23$
- $= 3 \times 4 \times 5 \bmod 23 = 60 \bmod 23 = 14$

Euler's Theorem

- Euler's theorem can be thought of as a generalization of Fermat's little theorem.
- The modulus in the Fermat theorem is a prime, the modulus in Euler's theorem is an integer.
- We introduce two versions of this theorem.
- The first version is
- If a and n are coprime, then $a^{\varphi(n)} \equiv 1 \pmod{n}$.
- It means, If power of a is $\varphi(n)$ then its remainder is 1.
- Where $\varphi(n)$ is totient function.

Euler's Theorem

- Find the result of $6^{24} \bmod 35$.
- Here 6 and 35 are co-prime.
- So, $6^{24} \bmod 35 = 6^{\phi(35)} \bmod 35$
- $\phi(35) = \phi(7 \cdot 5) = 6 \cdot 4 = 24$
- 1

Euler's Theorem

- The second version of Euler's theorem removes the condition that a and n should be coprime.
- $a < n$, then
- $a^{\varphi(n) + 1} \equiv a \pmod{n}$.
- Find the result of $20^{62} \pmod{77}$.
- $\varphi(77) = \varphi(11 \cdot 7) = 10 \cdot 6 = 60$
- $20^{62} \pmod{77} = 20^{\varphi(77)+1} \cdot 20 \pmod{77}$
- $[(20^{\varphi(77)+1} \pmod{77}) \cdot (20 \pmod{77})] \pmod{77}$
- $[20 \cdot 20] \pmod{77}$
- $= 15$

Euler's Theorem

- The second version of Euler's theorem removes the condition that a and n should be coprime.
- If $n = p \times q$, $a < n$, and k an integer, then
- $a^{\varphi(n) + 1} \equiv a \pmod{n}$.
- Find the result of $20^{62} \bmod 77$.
- $\varphi(77) = \varphi(11 \times 7) = 10 \times 6 = 60$
- $20^{62} \bmod 77 = 20^{\varphi(77)+1} * 20 \bmod 77$
- $[(20^{\varphi(77)+1} \bmod 77) * (20 \bmod 77)] \bmod 77$
- $[20 * 20] \bmod 77$
- $= 15$

Euler's Theorem

- We can use Euler's theorem to calculate multiplicative inverse.
- $a \times a^{-1} \bmod n$
- $= a \times a^{-1} \times 1 \bmod n$
- $= [(a \times a^{-1} \bmod n) (1 \bmod n)] \bmod n$
- $= [(a \times a^{-1} \bmod n) (a^{\varphi(n)} \bmod n)] \bmod n$
- $= a \times a^{-1} \times a^{\varphi(n)} \bmod n$
- $a \times a^{\varphi(n) - 1} \bmod n$
- $= a^{\varphi(n)} \bmod n$
- $= 1$

Euler's Theorem

- Find $60^{-1} \bmod 187$
- $60^{-1} \bmod 187$
- $=60^{\phi(187)-1} \bmod 187$
- $\phi(187)=\phi(11*17)=10*16=160$
- Now let come back
- $60^{\phi(187)-1} \bmod 187$
- $=60^{159} \bmod 187$
- $a=60, m=159, n=187, z=1$

Euler's Theorem

- $a=60$, $m=159$, $n=187$, $z=1$

i	b_i of m	$z=z^2 \bmod n$	$z=z*a \bmod n$
7	1	$1^2 \bmod 187=1$	$1*60 \bmod 187=60$
6	0	$60^2 \bmod 187=47$	$=47$
5	0	$47^2 \bmod 187=152$	$=152$
4	1	$152^2 \bmod 187=103$	$=103*60 \bmod 187=9$
3	1	$9^2 \bmod 187=81$	$=81*60 \bmod 187=185$
2	1	$185^2 \bmod 187=4$	$=4*60 \bmod 187=53$
1	1	$53^2 \bmod 187=4$	$=4*60 \bmod 187=53$
0	1	$53^2 \bmod 187=4$	$=4*60 \bmod 187=53$

Miller Robin's Primality Test

- Miller Robin's primality test is based on fermat's little theorem.
- For a prime p and any positive integer a following is true
- $a^{p-1} \equiv 1 \pmod{p}$.
- Here p does not divide a .
- It generates a positive integer a such that $1 < a < p-1$
- Test p using miller robin's primality test with chosen a .
- Miller robin's test can return sure composite or probable prime

Miller Robin's Primality Test

- If it returns probable prime, then generate a for sufficient number of time s and test p for primality again.
- If it returns probable prime for s times then we can say p is prime with probability $1-1/4^s$
- For $s=10$, probability > 0.9999

Miller Robin's Primality Test

- Test_prime(){
- INPUT n
- IF $n \leq 1$, return “neither prime nor composite”
- ELSE IF $n < 3$ return “Prime”
- ELSE IF n is even, return “Composite”
- Choose sufficiently large s
- FOR i= 1 to s
- Generate random integer $1 < a < n-1$
- If miller_rabin_test(n,a)=“composite”, return “composite”
- END FOR
- RETURN “prime”}

Miller Robin's Primality Test

- miller_rabin_Test(n, a) {
- Find an positive integer k and positive odd integer m such that $n-1 = m \times 2^k$
- $t = a^m \bmod n$
- IF $t=1$ or $t=n-1$, return “probable prime”
- FOR $i = 1$ to $k-1$
- $t = t^2 \bmod n$
- If $t=n-1$ return “probable prime”
- END FOR
- RETURN “composite” }

Miller Robin's Example

- Lets test $n=561$
- $561-1=m \times 2^k$
- Or, $m=560/2^k$
- Put $k=1$, $m=560/2=280$
- Put $k=2$, $m=560/4=140$
- Put $k=3$, $m=560/8=70$
- Put $k=4$, $m=560/16=35$
- Put $k=5$, $m=560/32=17.5$
- So, $k=4$ and $m=35$

Miller Robin's Example

- Let randomly generated integer be $a=2$
- $T = 2^{35} \bmod 561 = 263$

i	t
1	$263^2 \bmod 561 = 166$
2	$166^2 \bmod 561 = 67$
3	$67^2 \bmod 561 = 1$

- So, 561 is composite

Miller Robin's Example

- Lets test $n=29$
- $29-1=7 \times 2^2$, which means $m=7$ and $k=2$
- Let randomly generated integer be $a=10$
- $t = 10^7 \bmod 29 = 10^2 \cdot 10^2 \cdot 10^2 \cdot 10 \bmod 29 = 17$

i	$t = t^2 \bmod n$
1	$17^2 \bmod 29 = 28$

- So, 29 is probable prime

Miller Robin's Example

- Lets test $n=29$
- $29-1=7 \times 2^2$, which means $m=7$ and $k=2$
- Let randomly generated integer be $a=3$
- $t = 3^7 \bmod 29 = 3^4 \cdot 3^3 \bmod 29 = 23 \cdot 27 \bmod 29 = 12$

i	$t = t^2 \bmod n$
1	$12^2 \bmod 29 = 28$

- So, 29 is still probable prime

Primitive root

- A number a is a primitive root modulo n if every number co-prime to n is congruent to power of a modulo n .
- The a is said to be primitive root of n if $a \bmod n$, $a^2 \bmod n$, $a^3 \bmod n$,, $a^{n-1} \bmod n$ are distinct.
- Is 2 a primitive root of 5?
- $2^1 \bmod 5 = 2$
- $2^2 \bmod 5 = 4$
- $2^3 \bmod 5 = 3$
- $2^4 \bmod 5 = 1$
- So, 2 is primitive root of 5

Logarithm

- It is always easy to compute exponent of a number powered with some number.
- $2^5=32$, $2^{100}=1267650600228229401496703205376$
- How to compute the exponent given its base and result.
- For example, $2^x=32$, here $x=5$
- $3^x=129140163$
- Computing exponent satisfying some relation is difficult for large values.
- We can use logarithm in such case.
- $x=\log_3(129140163)=17$

Discrete Logarithm

- In similar way, we can use logarithm to find the exponent in modular arithmetic.
- In such case, logarithm is called discrete logarithm since it always provides discrete values.
- It is represented with $\text{dlog}_{a,p}$
- Here, a is base and p is modulus.
- For example
- $b = a^i \bmod p$
- $i = \text{dlog}_{a,p}(b) = \log_a b \bmod p$
- Computing discrete logarithm is difficult as we need to try all the powers.

Discrete logarithm

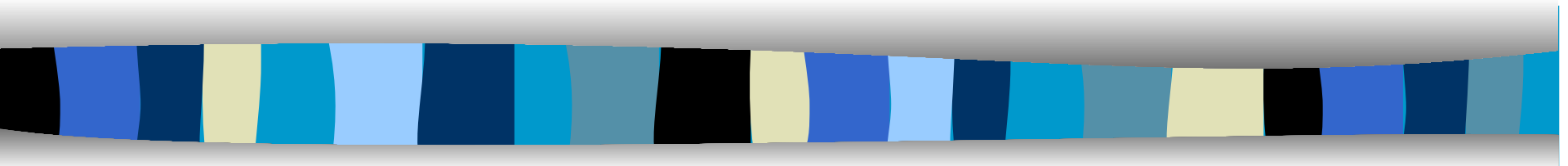
- So, we can use this property for cryptography.
- It will be more difficult if discrete logarithm provides unique value.
- This is only possible if the base a is primitive root of p .
- Every discrete logarithm may not have solutions.
- $x = \log_3 4 \bmod 13$ has no solution as 3 is not primitive root of 13.
- The equation $b = a^i \bmod p$ has only one solution if b is relative prime to p and a is primitive root to p .
- $i = \text{dlog}_{a,p}(b)$

Discrete logarithm

- Find x in each of the following cases:
 - a. $4 \equiv 3^x \pmod{7}$.
 - b. $6 \equiv 5^x \pmod{7}$.
- Solution
- We can easily use the tabulation of the discrete logarithm in Table 9.6.
- a. $4 \equiv 3^x \pmod{7} \rightarrow x = L_3 4 \pmod{7} = 4 \pmod{7}$
- b. $6 \equiv 5^x \pmod{7} \rightarrow x = L_5 6 \pmod{7} = 3 \pmod{7}$

y	1	2	3	4	5	6
$x = L_3 y$	6	2	1	4	5	3
$x = L_5 y$	6	4	5	2	1	3

Asymmetric Cryptography



RSA, Elgamal

Problems with Symmetric Cryptography

- Secret key must be shared among communicants through secure medium such as the use of a key distribution center.
- The sender cannot sign the document digitally with symmetric cryptography.
- To solve these problems, we have asymmetric cryptography.

Misconception on Symmetric Crypto

- First misconception is that public-key encryption is more secure from cryptanalysis than is symmetric encryption.
- In fact, the security of any encryption scheme depends on the length of the key.
- A second misconception is that public-key encryption has made symmetric encryption obsolete.
- However, asymmetric has more computational overhead.

Misconception on Symmetric Crypto

- Some applications of symmetric cryptography needs central agent to manage keys which is not simpler and efficient than symmetric cryptography.

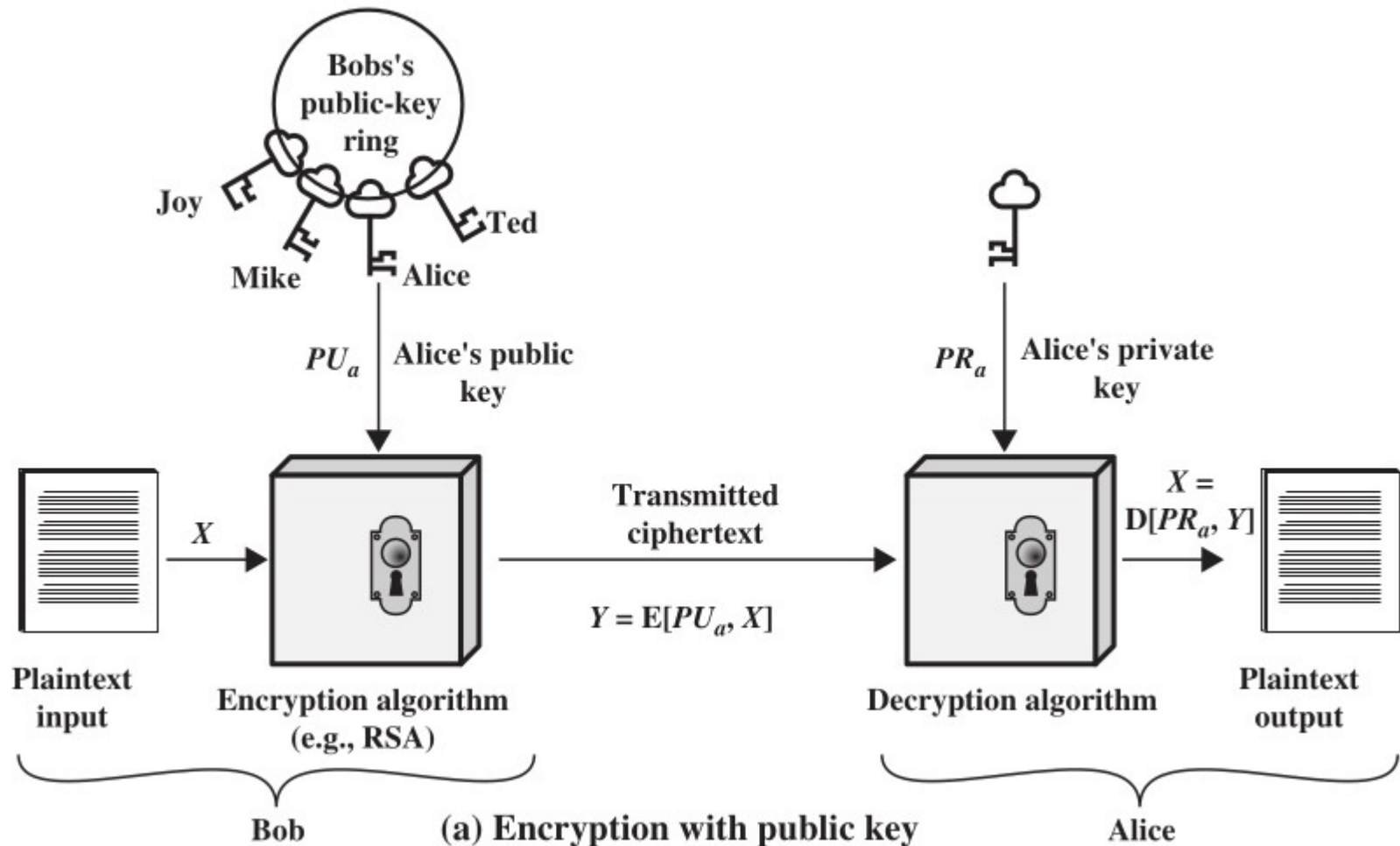
Asymmetric Cryptography

- In Asymmetric cryptography, each user generates a pair of keys to be used for the encryption and decryption of messages.
- Each user shares public key through in a public register or other accessible file.
- The companion key is kept private.
- Each user maintains a collection of public keys obtained from others.
- If Bob wishes to send a confidential message to Alice, Bob encrypts the message using Alice's public key.

Asymmetric Cryptography

- When Alice receives the message, she decrypts it using her private key.
- No other recipient can decrypt the message because only Alice knows Alice's private key.
- So, asymmetric cryptography uses different keys for encryption and decryption.
- The use of two keys has profound consequences in the areas of confidentiality, key distribution, and authentication

Asymmetric Cryptography



Public Key Cryptography

- Asymmetric cryptography is also known as public key cryptography.
- It is because Asymmetric cryptography publicly shares the public key.
- The development of public-key cryptography is the greatest revolution in the entire history of cryptography.
- Public-key algorithms are based on mathematical functions rather than on substitution and permutation.

Characteristics of public crypto

- It is computationally infeasible to determine the decryption key given only knowledge of the cryptographic algorithm and the encryption key.
- Either of the two related keys can be used for encryption, with the other used for decryption.

Ingredients of public cryptography

- **Plaintext:** This is the readable message or data that is fed into the algorithm as input.
- **Encryption algorithm:** The encryption algorithm performs various transformations on the plaintext.
- **Public and private keys:** This is a pair of keys that have been selected so that if one is used for encryption, the other is used for decryption.
- **Ciphertext:** This is the scrambled message produced as output.
- **Decryption algorithm:** produces the original plaintext using the matched key.

Applications of public cryptography

- **Encryption/decryption:** The sender encrypts a message with the recipient's public key.
- **Digital signature:** The sender “signs” a message with its private key. Signing is achieved by a cryptographic algorithm applied to the message or to a small block of data that is a function of the message.
- **Key exchange:** Two sides cooperate to exchange a session key. Several different approaches are possible, involving the private key(s) of one or both parties.

Applications of public cryptography

- Some public cryptography with their applications
- Digital Signature Standard

Algorithm	Encryption/Decryption	Digital Signature	Key Exchange
RSA	Yes	Yes	Yes
Elliptic Curve	Yes	Yes	Yes
Diffie-Hellman	No	No	Yes
DSS	No	Yes	No

Requirements for Public Crypto

- It is computationally easy for to generate a pair (public key Pub, private key PRb).
- It is computationally easy for a sender A, knowing the public key and the message to be encrypted, M, to generate the corresponding ciphertext:
- $C = E(\text{PUb}, M)$
- It is computationally easy for the receiver B to decrypt the ciphertext using the private key
- $M = D(\text{PRb}, C) = D[\text{PRb}, E(\text{PUb}, M)]$

Requirements for Public Crypto

- It is computationally infeasible for an adversary, knowing the public key, Pub , to determine the private key, PRb .

Cryptanalysis of Public Key Crypto

- Use of large key in public cryptography to make brute force attack infeasible makes it very slow.
- So, public cryptography is confined to key management and signature applications.
- Computing private key given public key is infeasible.
- If DES secret key is being shared using public cryptography then, attacker can encrypt all possible 56 bit keys using public key.
- If encryption matched with the ciphertext then attacker will know the DES key.

RSA

- The most common public-key algorithm is the RSA cryptosystem, named for its inventors (Rivest, Shamir, and Adleman).
- RSA uses two exponents, e and d , where e is public and d is private.
- Suppose P is the plaintext and C is the ciphertext. Alice uses $C = P^e \bmod n$ to create ciphertext C from plaintext P . Here e is public key of Bob.
- Bob uses $P = C^d \bmod n$ to retrieve the plaintext sent by Alice. Where d is private key of Bob.

RSA

- The modulus n , a very large number, is created during the key generation process.
- Encryption and decryption use modular exponentiation.
- Exponentiation takes $O((\log n)^3)$ operations (easy)
- For decryption, attacker must know private key d .
- Which is one of the factor of n .
- Factorization takes $O(e \log n \log \log n)$ operations (hard)

RSA: Key generation

- Select large distinct primes p and q using primality test.
- $n=pq$
- $\phi(n)=(p-1)(q-1)$
- Select positive integer e such that $\gcd(e, \phi(n))=1$
- Select d such that $ed \equiv 1 \pmod{\phi(n)}$
- Public key $\{e, n\}$
- Private key $\{d, n\}$

RSA: Encryption

- m is the plaintext
- Value of m is between 0 to $n-1$
- $C = m^e \bmod n$

RSA: Decryption

- C is the plaintext
- Value of C is between 0 to $n-1$
- $m = C^d \bmod n$

RSA: Correctness of Decryption

- $C^d \bmod n$
- $= (m^e)^d \bmod n$
- $= m^{ed} \bmod n$
- $= m^{q\phi(n)+1} \bmod n$
- $= m^{q\phi(n)} \times m \bmod n$
- $= [(m^{\phi(n)} \bmod n)^q (m \bmod n)] \bmod n$
- $= [1^q (m \bmod n)] \bmod n$
- $= [1 m] \bmod n = m$

RSA example: Key Generation

- Select $p=17$, $q=11$
- $n=pq=17 \times 11=187$
- $\phi(n)=(p-1)(q-1)=16 \times 10=160$
- Select $e=7$ as $\gcd(\phi(187), 7)=1$
- Compute inverse of e using extended euclidean algorithm
- $ed \equiv 1 \pmod{\phi(n)}$

RSA example: Key Generation

q	n	b	r	t1	t2	$t=t1-t2*q$
22	160	7	6	0	1	-22
1	7	6	1	1	-22	23
6	6	1	0	-22	23	-160
	1	0		23	-160	

- So, multiplicative inverse of e (7) is $d(23)$
- Private key is $\{23, 187\}$
- Public key is $\{7, 187\}$

RSA example: Encryption

- Let plaintext, $m=118$, $e=7$, $n=187$
- $C=m^e \bmod n = 118^7 \bmod 187$
- Here, 7 in binary is 111 which is 4,2,1
- $=118^4 \times 118^2 \times 118 \bmod 187$
- $118^2 \bmod 187 = 13924 \bmod 187 = 86$
- $118^4 \bmod 187 = (118^2)^2 \bmod 187 = (118^2 \bmod 187)^2 \bmod 187$
- $= 86^2 \bmod 187 = 7396 \bmod 187 = 103$
- Now, $118^4 \times 118^2 \times 118 \bmod 187 = 103 \times 86 \times 118 \bmod 187$
- $= 8858 \times 118 \bmod 187 = 69 \times 118 \bmod 187 = 8142 \bmod 187 = 101$

RSA example: Decryption

- Ciphertext is $C=101$, $d=23$, $n=187$
- $m=C^d \bmod n = 101^{23} \bmod 187$
- 23 in binary is 10111 which is 16,4,2,1
- $101^{23} \bmod 187 = 101^{16} \times 101^4 \times 101^2 \times 101 \bmod 187$
- $= 101^2 \bmod 187 = 10201 \bmod 187 = 103$
- $= 101^4 \bmod 187 = (101^2)^2 \bmod 187 = (101^2 \bmod 187)^2 \bmod 187$
- $= 103^2 \bmod 187 = 10609 \bmod 187 = 137$
- $= 101^8 \bmod 187 = (101^4)^2 \bmod 187 = (101^4 \bmod 187)^2 \bmod 187$
- $= 137^2 \bmod 187 = 18769 \bmod 187 = 69$

RSA example: Decryption

- $101^{16} \bmod 187 = (101^8)^2 \bmod 187 = (101^8 \bmod 187)^2 \bmod 187$
- $= 69^2 \bmod 187 = 4761 \bmod 187 = 86$
- Now, $101^{23} \bmod 187 = 101^{16} \times 101^4 \times 101^2 \times 101 \bmod 187$
- $= 86 \times 137 \times 103 \times 101 \bmod 187$
- $= [(86 \times 137 \bmod 187) \times (103 \times 101 \bmod 187)] \bmod 187$
- $= 1 \times 118 \bmod 187$
- $= 118$

Elgamal Cryptosystem

- Elgamal is another public-key cryptosystem.
- It is named after the inventor, Taher ElGamal.
- ElGamal is based on the discrete logarithm problem.

Elgamal Encryption

- 1. Select large prime p and its primitive root e_1 in group $G = \langle \mathbb{Z}_{p^*}, \times \rangle$
- 2. Generate random integer d such that $1 \leq d \leq p-2$
- 3. Compute $e_2 = e_1^d \bmod p$
- 4. Select random integer r such that $1 < r < p-1$
- 4. Now compute initial cipher text c_1
 $c_1 = e_1^r \bmod p$
- 5. Now compute actual cipher text c_2
 $c_2 = m \times e_2^r \bmod p$

Elgamal: Decryption

- $m = [c_2(c_1^d)^{-1}] \bmod p$
- Correctness
- $[C_2 \times (C_1^d)^{-1}] \bmod p$
- $= [(e_2^r \times m) \times (e_1^{rd})^{-1}] \bmod p$
- $= (e_1^{dr}) \times m \times (e_1^{rd})^{-1} \text{ as } e_2 = e_1^d \bmod p$
- $= m$
- Elgamal uses the property that given e_2 , e_1 , and p , it is infeasible to calculate $r = \text{dlog}_{e_1} e_2 \bmod p$

Elgamal Example: Key Generation

- Lets use Z_{19} , $p=19$, $e_1=10$, $d=5$
- $e_2=e_1^d \bmod 19$
- $=10^5 \bmod 19$
- $=3$
- Private key $d=5$
- Public key $e_1=10$, $e_2=3$, $p=19$

Elgamal Example: Encryption

- Public key $e_1=10$, $e_2=3$, $p=19$, $m=17$
- Select random integer $r=6$ in Z_{19}
- $c_1=e_1^r \bmod p$
- $=10^6 \bmod 19=11$
- $c_2=m \times e_2^r \bmod p$
- $=17 \times 3^6 \bmod 19$
- $=17 \times 7 \bmod 19$
- $=5$

Elgamal Example: Decryption

- private key $d=5$, $c_1=11$, $c_2=5$, $p=19$
- $m=c_2 \times (c_1^d)^{-1} \bmod p$
- $=5 \times (11^5)^{-1} \bmod 19$
- $=[5 \times (11^5 \bmod 19)^{-1}] \bmod 19$
- 5 in binary is 101 which is 4, 1
- $[5 \times (11^4 \times 11 \bmod 19)^{-1}] \bmod 19$
- $11^2 \bmod 19=7$
- $11^4 \bmod 19=(11^2)^2 \bmod 19=7^2 \bmod 19=11$

Elgamal Example: Decryption

- Now, $[5 \times (11^4 \cdot 11 \bmod 19)^{-1}] \bmod 19$
- $= [5 \times (11 \times 11 \bmod 19)^{-1}] \bmod 19$
- $= [5 \times (7)^{-1}] \bmod 19$
- $= 5 \times 11 \bmod 19 = 17$

Distribution of Secret Key using Public cryptography

- Two communicant must share the secret key before they actually communicate using symmetric cryptography.
- We can use public key cryptography to share the secret key of symmetric cryptography.
- One such method is Diffie-Hellman protocol.

Diffie-Hellman Protocol

- Two communicant must share the secret key before they actually communicate using symmetric cryptography.
- We can use public key cryptography to share the secret key of symmetric cryptography.
- One such method is Diffie-Hellman protocol.
- In the Diffie-Hellman protocol two parties choose two numbers p and g .
- The first number p is a large prime number on the (1024 bits).
- The second number, g , is a generator of order $p - 1$ in the group $\langle \mathbb{Z}_p^*, \times \rangle$.

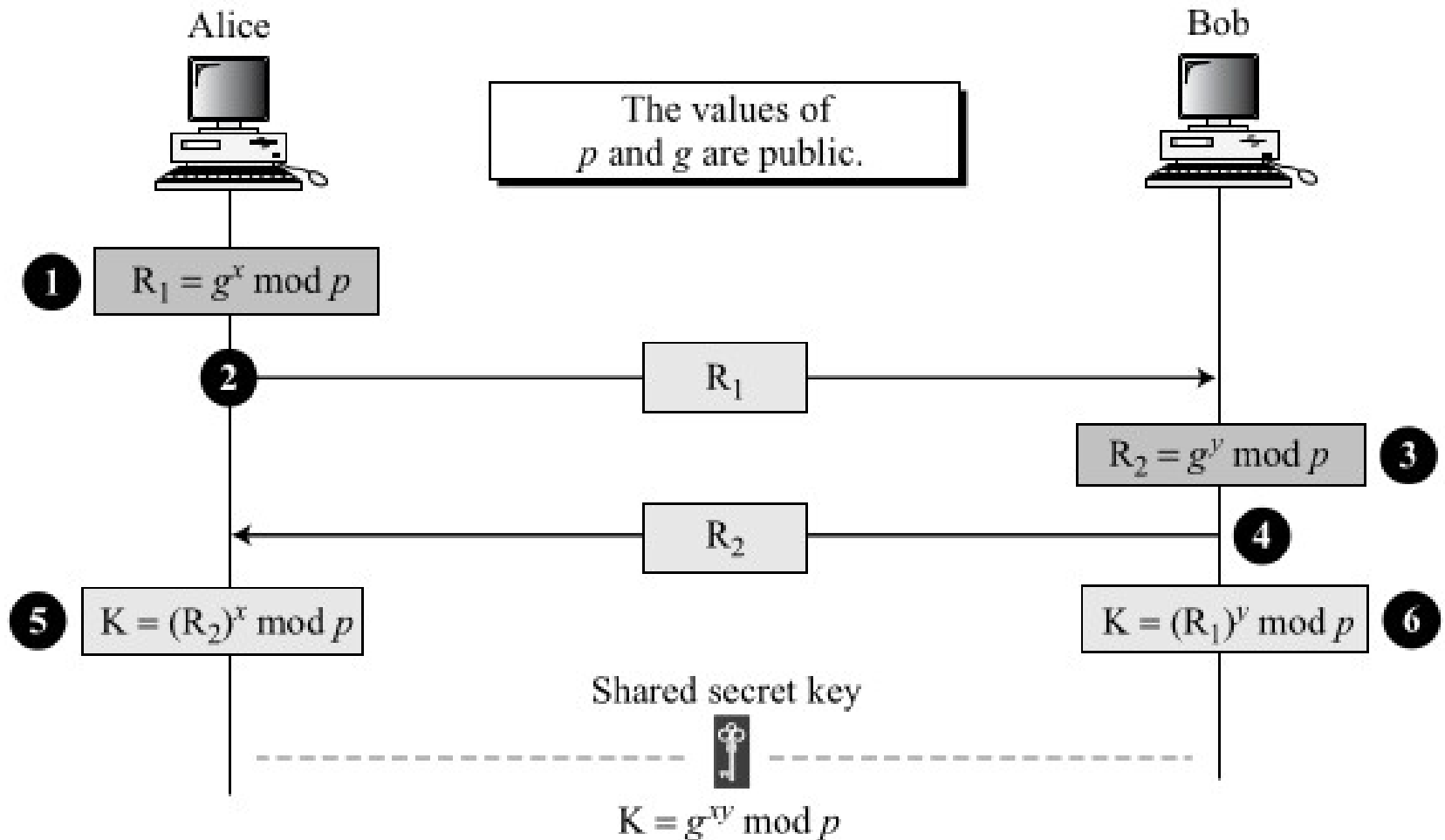
Diffie-Hellman Protocol

- Suppose Alice and Bob are communicating each other.
- 1. Alice and Bob agrees upon a large prime number p and a prime number g .
- 2. Alice chooses a large random number x such that $0 \leq x \leq p-1$ and calculate $R_1 = g^x \bmod p$.
- 3. Bob also chooses another large random number y such that $0 \leq y \leq p-1$ and calculate $R_2 = g^y \bmod p$.

Distribution of Secret Key using Public cryptography

- 5. Alice sends R_1 to Bob.
- 6. Bob sends R_2 to Alice.
- 7. Alice calculates shared key, $K = (R_2)^x \bmod p$.
- 8. Bob also calculates shared key, $K = (R_1)^y \bmod p$.
- Both K is same because
- $K = (R_2)^x \bmod p = (g^y \bmod p)^x \bmod p = g^{yx} \bmod p$

Distribution of Secret Key using Public cryptography



Diffie Hellman Example

- Alice and Bob, $p=19$, $g=5$
- Alice, $x=6$
- Alice, $R_1 = g^x \bmod p = 5^6 \bmod 19 = 7$
- Bob, $y=7$
- Bob, $R_2 = g^y \bmod p = 5^7 \bmod 19 = 16$
- Alice Key, $K = R_2^x \bmod p = 16^6 \bmod 19 = 7$
- Bob key, $K = R_1^y \bmod p = 7^7 \bmod 19 = 7$

Man in the middle attack

- 1. Alice chooses x , calculates $R1 = g^x \bmod p$, and sends $R1$ to Bob.
- 2. Eve, the intruder, intercepts $R1$. She chooses z , calculates $R2 = g^z \bmod p$, and sends $R2$ to both Alice and Bob.
- 3. Bob chooses y , calculates $R3 = g^y \bmod p$, and sends $R3$ to Alice.
- $R3$ is intercepted by Eve and never reaches Alice.
- 4. Alice and Eve calculate $K1 = g^{xz} \bmod p$, which becomes a shared key between Alice and Eve.

Man in the middle attack

- Alice, however, thinks that it is a key shared between Bob and herself.
- 5. Eve and Bob calculate $K2 = g^{zy} \bmod p$, which becomes a shared key between Eve and Bob.
- Bob, however, thinks that it is a key shared between Alice a

Man in the middle attack

